Q-1: Given cache size = 4 blocks, Block Address = 129;
Find the cache index where this block will be mapped?

Q-2: Find the cache block address for the memory address = 321, if the block size is 10 bytes.

Q-3: Find the cache index for the memory address = 239, given that the Block size is 32 bits and the cache size is 64 bits.

Q-4: Find the cache index, Tag, and offset bits for the memory address = 297, given that the Block size is 8 bits and the cache size is 32 bits.

Q-5:
Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	1	011	Data
011	1	001	Data
100	0		
101	1	000	Data
110	1	010	Data
111	0		

1. 100100
2. 111010
3. 010110
4. 100100
5. 111110

a) For the following memory access, determine if it will be a hit or miss.
b) Calculate the hit and miss ratio.
c) Find the AMAT if the hit time is 2s and the miss penalty is 5s.
d) If the CPU has a 2s clock, then determine how many cycles are required per instruction.